

Characteristics from a graph



1

- a** Yes **b** No

2

- a They become very large in the negative direction.
- b They become very large in the positive direction.

3

- a**
- i It has an absolute minimum.
 - ii Yes, there is a local minimum.
- b**
- i No, it does not have either.
 - ii Yes, it has local maximums and minimums.
- c**
- i Yes, it has an absolute maximum.
 - ii Yes, there are local maximums and minimums.

4

- a**

 - i $(3, 0)$
 - ii $(0, -3)$

b

 - i $(4, 0)$
 - ii $(0, 4)$

c

 - i $(-1, 0)$
 - ii $(0, -2)$

5

- a**
- i $(-8, 0), (0, 0)$
 - ii $(0, 0)$
 - iii They become very large in the positive direction
 - iv They become very large in the positive direction
- b**
- i $(-2, 0), (-4, 0), (3, 0)$
 - ii $(0, 5)$
 - iii They become very large in the negative direction
 - iv They become very large in the positive direction



c

- i The function does not have x -intercepts.
- ii $(0, 1)$
- iii They become very large in the positive direction
- iv They approach zero

6

- a The function does not have x -intercepts.
- b $(0, 1)$
- c They become very large in the positive direction

7 $(-2, 0)$

8

- a 0 km
- b 30 km
- c 1 hour

9 1 km

10 $\frac{3}{2} < x < 2$

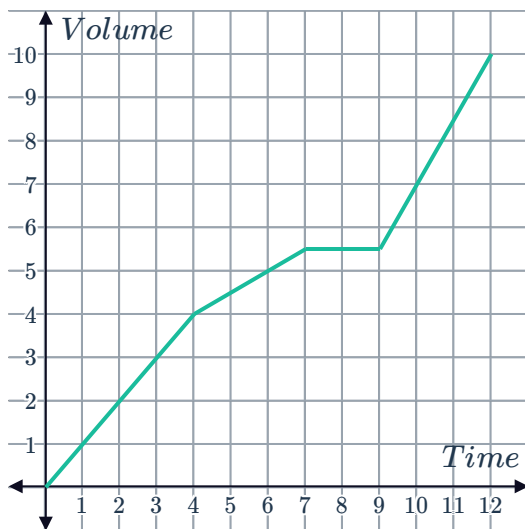
11

- a No
- b No
- c Yes
- d No

12 $5 \leq t \leq 6$

13

a



b

$\frac{5}{6}$ million gallons of water per hour

Additional questions

14

a

i Domain: $-2 < x \leq 4$ ii Range: $-5 < y \leq 1$ iii x -intercept is 3
 y -intercept is -3

iv Rate of change is 1

b

i Domain: all real values

ii Range: all real values

iii x -intercept is 3
 y -intercept is 6iv Rate of change is -2

c

i Domain: all real values

ii Range: $y = 2$ iii There is no x -intercept
 y -intercept is 2

iv Rate of change is 0

d

i Domain: $-2 \leq x \leq 4$ ii Range: $-2 \leq y \leq 1$ iii x -intercept is 2
 y -intercept is -1 iv Rate of change is $\frac{1}{2}$

15



- a $(0, 9)$
- b The height from which the soccer ball was kicked.
- c $(3, 0)$
- d The time at which the soccer ball hits the ground.